GCFormer: Global Context-Aware Transformer for Remote Sensing Image Change Detection

Wanting Yu, Li Zhuo, *Member, IEEE*, and Jiafeng Li, *Member, IEEE*

***Abstract*— In recent years, Transformer-based change detection (CD) in remote sensing images has achieved significant advances, making it an emerging hot research topic. However, the current CD methods suffer from some problems, such as incomplete detection of change regions and missed detection of small change regions. In this article, a global context-aware Transformer is proposed for CD tasks, named GCFormer, to address above issues by efficiently enhancing the global context information. It is fulfilled from two aspects based on the hybrid convolutional neural network (CNN) + Transformer framework. First, a multireceptive-field Conv-Attention (MRFCA) mechanism is designed, which combines dilated convolutions with multiple rates (DCMRs) and Conv-Attention, fully leveraging the advantages of convolution operation and self-attention mechanism. It is embedded at the highest layer of CNN to extract multireceptive-field global context information. Second, a context-aware relative position encoding (CRPE) mode is proposed to replace the absolute position encoding (APE) mode of Transformer. As a result, it can capture long-range dependency more efficiently and further enhance the global context information extraction and representation ability of the network. Experimental results on three public benchmark datasets of LEVIR-CD, WHU-CD, and DSIFN-CD show that, the proposed GCFormer achieves superior detection performance with lower model complexity than the state-of-the-art (SOTA) Transformer-based CD methods. The source code is available at https://github.com/yuwanting828/yuwanting828.github.io.**

***Index Terms*— Change detection (CD), global context-aware Transformer, long-range dependency, remote sensing images.**

I. Introduction

REMOTE sensing image change detection (RSICD) is to analyze, process, and compare multiple remote sensing images of the same area taken at different times to obtain information about changes of the land surface within a certain time [1]. It can be widely used in many applications, such as forest vegetation change monitoring, environmental dynamic monitoring, natural disaster assessment, urban governance, national land and resources survey, military defense, and so on. However, due to various complicated factors, including illumination, phenology, solar height and angle changes, diverse ground objects, and complex scenes, RSICD tasks enface great challenging.

In recent years, Transformer has achieved great success in the field of natural language processing (NLP) and has become an emerging hot research topic. The key to Transformer is the self-attention mechanism, which is capable of modeling long-range dependency. In addition, it employs multihead attention mechanism [2] to balance the potential bias that may be produced by using a single attention mechanism, which facilitates it to effectively integrate global semantic information. Due to the tremendous success of Transformer, researchers have applied it to visual tasks, such as object detection and semantic segmentation by combining it with convolutional neural network (CNN). This integration can fully exploit the advantages of CNN in extracting hierarchical local features and Transformer's ability of modeling the long-range dependency. DEtection TRansformer (DETR) [3] is a representative CNN + Transformer hybrid framework. Research results have shown that DETR can achieve better performance than the CNN-based methods, making it a new hotspot in the fields of visual tasks.

Based on the DETR framework, several RSICD methods have been proposed recently, such as MSPSNet, a novel deep multiscale Siamese network with a parallel convolutional structure (PCS) and self-attention [4], and bitemporal image Transformer (BIT) [5], which have significantly improved the detection accuracy. However, in terms of detection results, these methods still cannot deal with the change regions with drastic scale variations well, resulting in incomplete detection of change regions, and particularly missed detection for scattered small objects.

Global context information can effectively represent the relationship between different objects in an image. In tasks, such as object detection and semantic segmentation, effectively extracting and utilizing global context information facilitate to address the challenges related to drastic scale variations of the objects. Therefore, in this article, a global context-aware Transformer is proposed for RSICD, named GCFormer, which adopts a hybrid CNN + Transformer as the basic framework. GCFormer enhances the global context information from two aspects. First, a multireceptive-field Conv-Attention (MRFCA) mechanism is designed, which combines dilated convolutions with multiple rates (DCMRs) and Conv-Attention, fully leveraging the advantages of convolution operation and self-attention mechanism. It is embedded at the highest layer of CNN to extract multireceptive-field

Manuscript received 8 May 2023; revised 9 October 2023, 3 January 2024, and 9 March 2024; accepted 22 March 2024. Date of publication 25 March 2024; date of current version 5 April 2024. This work was supported in part by the Research and Development Program of Beijing Municipal Education Commission under Grant KM202210005007 and in part by Beijing Natural Science Foundation under Grant L211017. *(Corresponding author: Li Zhuo.)*

The authors are with Beijing Key Laboratory of Computational Intelligence and Intelligent System, Faculty of Information Technology, Beijing University of Technology, Beijing 100124, China (e-mail: yuwt@emails.bjut.edu.cn; zhuoli@bjut.edu.cn; lijiafeng@bjut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3381738

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

global context information. Second, a context-aware relative position encoding (CRPE) mode is proposed to replace the absolute position encoding (APE) mode of Transformer, so it can more efficiently capture long-range dependency, further enhancing the global context information extraction and representation ability of the network.

The main contributions of this article are as follows.

1) MRFCA is proposed and embedded at the highest layer of CNN, to extract multireceptive-field global context information. It facilitates to the complete detection of change regions with different scales, especially mitigating the missed detection of the scattered small regions.
2) CRPE mode is proposed to replace the APE mode of Transformer, which enables the network to capture long-range dependency more efficiently, thus further enhancing its ability to extract the global context information.

Experimental results on three publicly available benchmark datasets, LEVID-CD, WHU-CD, and DSIFN-CD, demonstrate that compared with the state-of-the-art (SOTA) RSICD methods, the proposed GCFormer can achieve a superior detection performance.

In the rest of this article, after reviewing related work in Section II, a detailed description of the proposed GCFormer is provided in Section III. Section IV reports the experimental results and analysis. Finally, the conclusions are drawn.

II. Related Work

With the rapid development of deep learning, significant achievements have been made on RSICD tasks. The existing methods can be roughly divided into two categories: CNN-based methods and Transformer-based methods. In this section, we will investigate the current research advances of these two kinds of RSICD methods.

A. CNN-Based RSICD Methods

It is well known that CNN can automatically learn powerful, universal, and hierarchical feature representation with the aid of large-scale manually annotated data samples. Therefore, CNN has been widely used in RSICD tasks and achieved superior detection performance [6], [7], [8].

In 2018, Daudt et al. [9] innovatively proposed the fully convolutional early fusion (FC-EF) method, which adopts a fully CNN framework for RSICD tasks and achieves a superior detection performance, far beyond that of the traditional change detection (CD) methods. On the basis of this work, Qu et al. [10] designed a dual-domain network (DDNet), which uses a splitting and fusing strategy to enhance the features in the center regions, further improving the detection performance.

However, it is difficult for the above methods to accurately distinguish the changing regions due to complex factors, such as noises, angle, and shadows. Siamese architecture can finely compare the similarity between two input images in the feature space and obtain discriminative feature representation, which is consistent with the goal of CD tasks. Therefore, researchers have applied the Siamese architecture to RSICD methods. They designed various attention mechanisms and feature fusion strategies based on this architecture, mainly focusing on feature learning and context information aggregation strategies. For example, Daudt et al. [9] proposed a skip connection method based on FC-EF, named fully convolutional Siamese concatenation (FC-Siam-Conc). Liu et al. [11] proposed a dual-task constrained deep Siamese convolutional network (DTCDSCN). Chen and Shi [12] proposed a spatial–temporal attention network (STANet). These methods alleviate false detection problems caused by inaccurate matching, through the context information aggregation strategies. For feature learning, Chen et al. [13] proposed a dual-attentive Siamese network (DASNet). Based on the standard encoder–decoder architecture. Fang et al. [14] designed a network structure named SNUNet. It combines Siamese network and NestedUNet as the backbone, which overcomes the problem of "pseudo-changes" and ensures the detection integrity of change regions.

The multiple downsampling operations adopted by CNN eventually cause the loss of global context semantic information, making CNN-based RSICD methods difficult to cope with drastic scale variations of change regions well. Especially, for scattered small objects, missed detection often occurs. Therefore, how to capture the global context information to keep the integrity of detected change regions has become a key issue that needs to be studied.

B. Transformer-Based RSICD Methods

Compared with CNN, which has limited receptive fields and mainly focuses on local features, Transformer uses a self-attention mechanism to model the long-range dependency. Unlike channel-attention and spatial-attention mechanisms, self-attention mechanism utilizes the intrinsic information of features for interaction, reducing the reliance on external information. Therefore, it is better at capturing the internal correlation of features and modeling long-range dependency. Researchers have designed various Transformer-based architectures [15], [16] for different computer vision tasks. The research results show that Transformer can expand the effective receptive fields and have stronger ability of modeling long-range dependency than CNN.

In recent years, several Transformer-based RSICD methods have emerged. Bandara and Patel [18] proposed ChangeFormer. It adopts a Transformer-based Siamese network architecture and can extract both coarse-grained and fine-grained features from a pair of coregistered images. However, ChangeFormer utilizes the vision Transformer [19] with high-computational complexity. To reduce the complexity, Zhang et al. [21] proposed SwinSUNet, a Siamese U-shaped network architecture, which uses the Swin Transformer [20] to parallelly process bitemporal images and extract multiscale features. The experimental results show that SwinSUNet is more effective in coping with change regions with drastic scale variations. Ke and Zhang [22] proposed a hybrid Transformer structure based on the token aggregation strategy, named H-TransCD. It captures the context information in bitemporal remote sensing images with a multiscale manner, thus enhancing the feature representation ability. Guo et al. [23]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

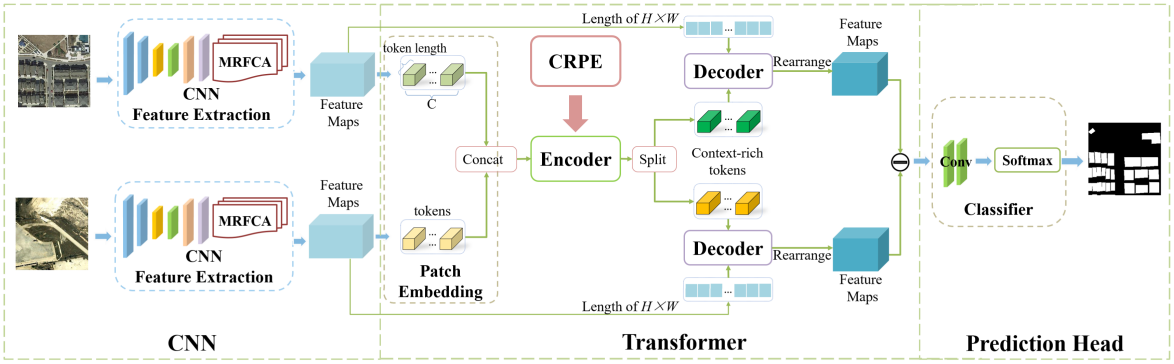


Fig. 1. Overall network framework of the proposed GCFormer.

proposed an iterative difference-enhanced Transformer (IDET) to optimize feature difference to highlight change regions and suppress invariant ones.

However, Transformer-based methods typically require high-computational costs and additional supervision. Also, due to the lack of inductive bias, more labeled data samples are required for network model training. As constructing large-scale RSICD dataset is time-consuming and labor-intensive, the advantages of Transformer cannot be fully utilized.

Therefore, researchers have proposed a hybrid network architecture that combines both CNN and Transformer to improve detection performance. Studies have shown that the combination of these two architectures can be more effective than only using single one. Chen et al. [5] proposed a hybrid CNN + Transformer RSICD method, which utilizes a deep feature difference framework and BIT to effectively model context information within the spatial–temporal domain. However, BIT only integrates CNN and Transformer, which may not capture the global context information effectively, resulting in missed detection of scattered small objects. Song et al. [24] proposed a multiscale Swin Transformer-based deeply supervised network (MSTDSNet) to refine multiscale features and enhance the feature distinguishability of change regions. Guo et al. [4] also proposed a multiscale Siamese network, named MSPSNet. It combines PCS and self-attention to extract multiscale features from bitemporal images. However, MSPSNet lacks spatial interaction information during global context modeling, which may limit its ability to capture effective location information during the encoding process, thus affecting the robustness of the model.

In this article, based on hybrid CNN + Transformer framework, GCFormer is proposed for RSICD tasks. It can enhance the global context information extraction and representation ability of the network. At the highest layer of the CNN, MRFCA is embedded to cope with the change regions with drastic scale variations. After that, the Transformer architecture with the proposed CRPE is used to encode and decode the global semantic feature extracted by the CNN, effectively enhancing the extraction and representation ability of global semantic features, thereby improving the detection performance.

III. Proposed RSICD Method

The overall framework of GCFormer proposed in this article is shown in Fig. 1. It adopts a hybrid architecture of CNN + Transformer, with ResNet18 serving as the CNN backbone.

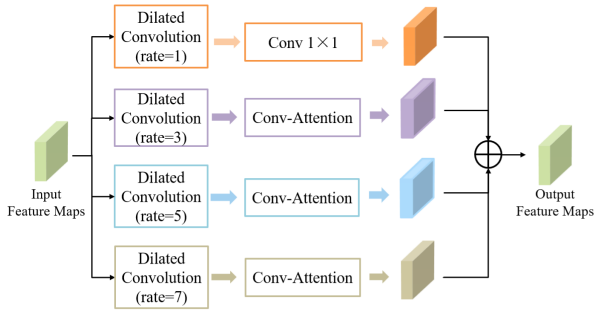


Fig. 2. Structure of the MRFCA module.

First, ResNet18 is adopted to extract the deep features from two input images. MRFCA is incorporated into the highest layer of ResNet18 to enhance the global context information. Subsequently, the enhanced features are fed into Transformer to generate the features with richer context information through the proposed CRPE. Finally, the difference maps are generated by subtracting the features of the two images, which are then input into a classifier, obtaining the output change map. The classifier contains two 3 × 3 convolutional layers and a softmax layer. The GCFormer mainly consists of two novel modules: MRFCA and CRPE. Their specific implementation details are introduced below.

A. Multireceptive-Field Conv-Attention

CNN stacks multiple convolutional and pooling layers to generate the hierarchical feature representation. The features at the higher layers typically contain global context information, which are suitable to locate the change regions correctly. However, the consecutive downsampling operations will result in the loss of global context information. So, incomplete detection of change regions and particularly missed detection for scattered small objects often occur in RSICD tasks.

To solve the problem, in this article, MRFCA is proposed to enhance the global context information. The structure of MRFCA is shown in Fig. 2. It consists of two key components, i.e., DCMR of 1, 3, 5, and 7 and Conv-Attention mechanism.

It is well known that dilated convolution can effectively enlarge receptive field without additional cost. So, MRFCA is designed to adopt parallel dilated convolutions with different rates to obtain multireceptive-field global context information. In addition, self-attention mechanism is good at modeling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

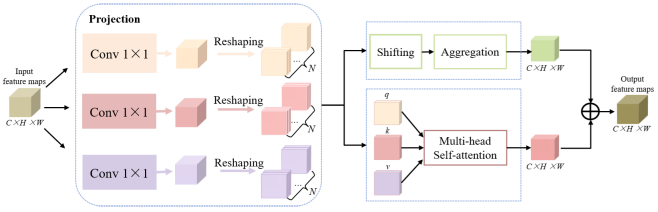


Fig. 3. Diagram of Conv-Attention mechanism.

the long-range dependency, which applies a weighted average operation based on the context of input features. The attention weights are dynamically computed via a similarity matching function between related pixel pairs, enabling the model to adaptively focus on different regions and capture the global context information. Therefore, in MRFCA, each dilated convolution branch is then followed by a Conv-Attention mechanism to further capture the global context information. The output features of each branch are fused together to yield the output features that contain richer global context information.

Conv-Attention was proposed in [17], in which the 3×3 convolution operation and self-attention mechanism are integrated into a unified framework. So, it can enjoy the benefits of both self-attention and 3×3 convolution operation. The experimental results demonstrate that Conv-Attention can achieve a better performance than single self-attention mechanism with lower computational cost [17].

The diagram of Conv-Attention mechanism is illustrated in Fig. 3. First, the input feature maps with a size of $C \times H \times W$ are projected with three 1×1 convolutions. Here, C represents the number of feature channels, and H and W denote the height and width of the feature maps, respectively. The projected features are reshaped into N groups, respectively, obtaining the intermediate features containing $3 \times N$ feature maps. Then, they are processed with different manners. One is to perform with the convolution manner through shifting and aggregation operations. The other is to perform with the self-attention manner, in which three feature maps serve as q, k, and v, respectively, following the traditional multihead self-attention calculations.

Finally, the output feature maps of the convolution and self-attention are pixelwise added together to obtain the final output feature maps, which contain richer global context information.

MRFCA is embedded into the highest layer of CNN. By MRFCA, the global context information can be enhanced, which facilitates to cope with the change regions with drastic scale variations well. Thereby, the detection performance can be improved.

B. Context-Aware RPE

As mentioned before, the combination of CNN and Transformer can extract and characterize the global–local information of the image, which facilitates to obtain superior detection performance. So, as shown in Fig. 1, the features of CNN are input into the Transformer, and the output of the Transformer is the enhanced feature maps, containing richer global context information.

The Transformer contains three key components, i.e., patch embedding, encoder, and decoder. The self-attention mechanism in the encoder is the most crucial part. It achieves its functionality by solely calculating the similarity between tokens, but ignores the position information of the tokens, which is important for RSICD tasks. Therefore, how to incorporate the position information into the encoder is critical while applying Transformer to RSICD tasks.

To address this issue, the existing Transformer-based RSICD methods commonly adopt an APE mode. The APE vector set $\mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_n)$ is simply added with the token embedding set $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)$, obtaining the APE $\boldsymbol{\delta}_i$, shown as follows:

$$\boldsymbol{\delta}_i = \mathbf{x}_i + \mathbf{p}_i. \tag{1}$$

Obviously, the APE mode does not consider the relative position relationship between the tokens, making the model unable to learn the context information well.

Therefore, in this article, a CRPE mode is proposed to replace the APE mode of Transformer. Different from APE, CRPE emphasizes the relative position information between tokens. The flowchart of the proposed CRPE is shown in Fig. 4. It is implemented through incorporating context mode into relative position encoding (RPE). So, in the following part, RPE is described first and followed by the context mode. They are described in detail below.

1) Relative Position Encoding: In self-attention, for the input sequence $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)$ containing n elements and the output sequence $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n)$, $\mathbf{z}_i \in \mathbb{R}^{d_z}$, each output element $\mathbf{z}_i$ is computed as a weighted sum of input elements

$$\mathbf{z}_i = \sum_{j=1}^{n} \alpha_{ij} \left(\mathbf{x}_j \mathbf{W}^V \right). \tag{2}$$

The calculation of each weight coefficient α_{ij} can be formulated as follows:

$$\alpha_{ij} = \text{softmax}\left(\frac{\left(\mathbf{x}_i \mathbf{W}^Q \right) \left(\mathbf{x}_j \mathbf{W}^K \right)^{\mathrm{T}}}{\sqrt{d_z}} \right) \tag{3}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ represent the elements at the positions of i and j in the sequence of tokens, respectively. The projection matrices $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$ are parameter matrices, which will multiply with elements $\mathbf{x}_i$ and $\mathbf{x}_j$ to generate the corresponding query ($\mathbf{q}$) and key ($\mathbf{k}$) vectors. In order to prevent overflow during softmax, the division by $\sqrt{d_z}$ is performed to normalize the variance.

In RPE, α_{ij} is computed as follows:

$$\alpha_{ij} = \text{softmax}\left(\frac{\left(\mathbf{x}_i \mathbf{W}^Q \right) \left(\mathbf{x}_j \mathbf{W}^K \right)^{\mathrm{T}} + \beta_{ij}}{\sqrt{d_z}} \right) \tag{4}$$

where β_{ij} is a learnable scalar, representing the relative position weight between the positions i and j.

2) Context Mode: For the self-attention mechanism, $\mathbf{q}$ determines the focus of attention, and the similarity between $\mathbf{q}$ and $\mathbf{k}$ helps the model to determine which position in the token sequence should be paid attention to. In this article,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

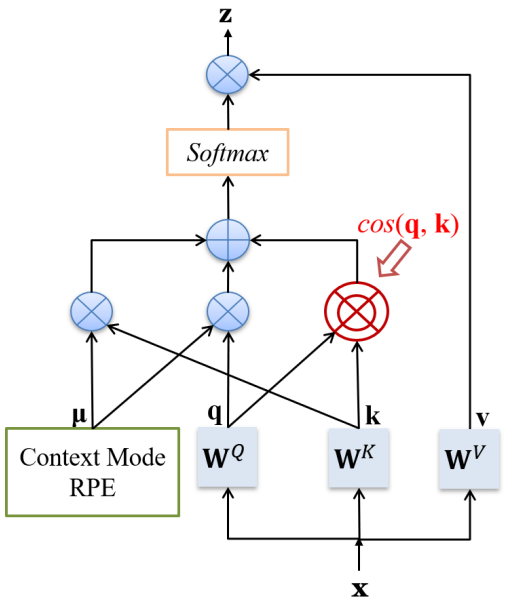


Fig. 4. Flowchart of the proposed CRPE mode.

context code is incorporated into RPE through β_{ij}, which can be computed as follows:

$$\beta_{ij} = \left(\mathbf{x}_i \mathbf{W}^Q\right)\left(\boldsymbol{\mu}_{ij}^K\right)^{\mathrm{T}} + \left(\mathbf{x}_j \mathbf{W}^K\right)\left(\boldsymbol{\mu}_{ij}^Q\right)^{\mathrm{T}} \tag{5}$$

where $\boldsymbol{\mu}_{ij}^K$ and $\boldsymbol{\mu}_{ij}^Q$ are learnable vectors. It can be seen that, β_{ij} not only calculates the relative position relationship between tokens of positions i and j, but also considers the interaction between the $\mathbf{q}$ vector and $\mathbf{k}$ vector during the encoding process, thereby utilizing more context information.

Currently, the dot product is commonly used to measure the similarity between the $\mathbf{q}$ and $\mathbf{k}$ in the self-attention mechanism. Also, the layer normalization operation is placed after the residual connection of each sublayer. However, these operations tend to the learned attention map being dominated by a small number of tokens and becoming relatively concentrated, which limits the representative ability of the global context information.

In this article, instead of the dot product operation, the cosine function is adopted in CRPE to compute the similarity between the $\mathbf{q}$ and $\mathbf{k}$. As the cosine function can yield the values within the range of 0–1, it ensures that the similarity score is distributed within the same range. This constraint prevents the learned attention map from being overly scattered, which could, otherwise, affect the detection accuracy. So, the formula for calculating α_{ij} in CRPE is

$$\alpha_{ij} = \mathrm{softmax}\left(\frac{\cos\left(\left(\mathbf{x}_i \mathbf{W}^Q\right), \left(\mathbf{x}_j \mathbf{W}^K\right)^{\mathrm{T}} + \beta_{ij}\right)}{\sqrt{d_z}}\right). \tag{6}$$

IV. Experimental Results and Analysis

To evaluate the effectiveness of the proposed method, we conducted the experiments on three benchmark RSICD datasets. The experimental results and analysis are introduced below.

A. Datasets and Evaluation Criteria

1) Datasets: Three public RSICD datasets used in the experiments include WHU-CD [25], LEVIR-CD [12], and DSIFN-CD [26].

WHU-CD is a publicly available building CD dataset released by Wuhan University in 2018. Each sample contains a pair of aerial images with a size of 32 207 × 15 354 pixels and a resolution of 0.2 m.

LEVIR-CD is a large-scale remote sensing building CD dataset released by Beihang University in 2020. It consists of 637 high-resolution Google Earth image patches, with a size of 1024 × 1024 pixels.

DSIFN-CD is a dataset released in 2020, which consists of 748 high-resolution Google Earth image pairs that provide pixel-level changes and annotations for land cover changes, such as roads, buildings, farmland, and water bodies.

In the experiments, each image is divided into nonoverlapping blocks with a size of 256 × 256 pixels. For fair comparison with other methods, the three datasets are divided into training, validation, and testing sets with the same pre-established splitting ratios. For WHU-CD, the ratios are 80%, 10%, and 10%, with 6096, 762, and 762 samples, respectively. Also, for LEVIR-CD and DSIFN-CD datasets, they are 70%, 10%, and 20%, with 7120, 1024, and 2048 samples, and 90%, 8%, and 2%, with 14 400, 1 360, and 192 samples, respectively.

2) Performance Evaluation Metrics: In this article, the most commonly used five metrics are adopted to evaluate the detection performance, namely, $F1$ score ($F1$), intersection over union (IoU), overall accuracy (OA), precision (Pre), and recall (Rec). The definitions of these metrics are as follows.

1) *Pre:* It represents the false alarm rate, which is the proportion of truly positive samples among all samples predicted as positive by the model. The formula is shown in the following equation:

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{7}$$

2) *Rec:* It represents the rate of missed detections, that is, how many of all positive samples are correctly judged as positive by the model. A higher recall indicates fewer false negatives. The formula is shown in the following equation:

$$\mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{8}$$

3) *F1:* It is the harmonic mean of precision and recall, which measures the balance between them. The formula is shown in the following equation:

$$F1 = \frac{2 * \mathrm{Pre} * \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}}. \tag{9}$$

4) *IoU:* It is used to measure of the overlapped ratio between the detected results and the ground truth (GT). The formula is shown in the following equation:

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{FP} + \mathrm{TP} + \mathrm{FN}}. \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PERFORMANCE COMPARISON RESULTS USING DIFFERENT CD METHODS ON THE LEVIR-CD DATASET

Method	Year	Architecture	F1	IoU	OA	Pre	Rec
SSGCN [28]	2020	GCN	/	/	92.80	/	73.21
MSGCN [29]	2021	GCN	/	/	95.68	/	76.11
DTCDSCN [11]	2020	SE-ResNet	87.67	78.05	98.77	88.53	86.83
STANet [12]	2020	ResNet18	87.26	77.40	98.66	83.81	91.00
IFNet [26]	2020	VGG16	88.13	78.77	98.87	94.02	82.93
SNUNet-CD [14]	2021	Siamese ConvNet	88.16	78.83	98.82	89.18	87.17
MSPSNet [4]	2022	SNN	89.18	/	/	91.38	87.08
MSTDSNet [24]	2022	Transformer+ResNet18	88.10	78.73	98.56	85.52	90.84
IDET [23]	2022	Transformer	88.90	80.20	98.10	91.30	86.60
BIT [5]	2022	Transformer+ResNet18	89.31	80.68	98.92	89.24	89.37
H-TransCD [22]	2022	Transformer	90.06	81.92	99.00	91.45	88.72
ChangeFormer [18]	2022	Transformer	90.40	82.48	99.04	92.05	88.80
GCFormer	2022	Transformer+ResNet18	90.83	83.21	99.08	89.34	92.50

TABLE II
PERFORMANCE COMPARISON RESULTS USING DIFFERENT CD METHODS ON THE WHU-CD DATASET

Method	Year	Architecture	F1	IoU	OA	Pre	Rec
DTCDSCN [11]	2020	SE-ResNet	71.95	56.19	97.42	63.92	82.30
STANet [12]	2020	ResNet18	82.32	69.95	98.52	79.37	85.50
IFNet [26]	2020	VGG16	83.40	71.52	98.83	96.91	73.19
SNUNet-CD [14]	2021	Siamese ConvNet	83.50	71.67	98.71	85.60	81.49
BIT [5]	2022	Transformer +ResNet18	83.98	72.39	98.75	86.64	81.48
ChangeFormer [18]	2022	Transformer	89.88	81.63	99.12	91.83	88.02
H-TransCD [22]	2022	Transformer	91.22	83.85	99.24	93.85	88.73
GCFormer	2022	Transformer+ResNet18	90.97	83.43	99.29	91.81	90.15

5) *OA:* It represents the percentage of correctly classified pixels in the CD map. A higher OA value indicates better performance. The formula is shown in the following equation:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FA} + \mathrm{TN} + \mathrm{FP}}. \tag{11}$$

B. Parameter Settings

In this article, the network has been constructed, trained, and tested on the PyTorch platform with the following hardware configuration: Intel[1] Core i5-4590 CPU at 3.30 GHz, and graphics card is NVIDIA GeForce GTX 1080 Ti. During network model training, the ResNet18 pretrained on ImageNet is adopted as the CNN backbone network, and the cross-entropy (CE) loss and SGD optimizer are used, with a learning rate of 0.01. The input image size is normalized to 256×256, the batch size is set to 8, and the epoch is 200. Data augmentation is performed during training by random flipping.

C. Comparison With the SOTA CD Methods

To verify the effectiveness of the proposed method in this article, we compared it with 12 SOTA RSICD methods, including CNN-based CD methods of DTCDSCN [11], STANet [12], IFNet [26], SNUNet-CD [14], graph convolutional network (GCN)-based [27] CD methods of semisupervised GCN (SSGCN) [28], multiscale GCN (MSGCN) [29], and Transformer-based CD methods of MSPSNet [4], deep difference network with BIT fusion [5], ChangeFormer [18], H-TransCD [22], IDET [23], and MSTDSNet [24].

The comparison results using different CD methods on the three datasets are shown in Tables I–III, and the optimal and suboptimal CD results are highlighted in red and blue, respectively.

The experimental data of other CD methods in the tables are quoted from the literature, and the symbol "/" denotes that the experimental data were not provided by the authors. For our GCFormer, the experiments are repeated ten times, and the average value is reported as the final performance. The optimal value of each metric in the remaining tables is highlighted in bold font.

It can be observed that in most cases, the proposed GCFormer can achieve the optimal or suboptimal results in terms of five evaluation metrics. In the following, we will provide the analysis of the experimental results of each dataset in detail.

1) LEVIR-CD Dataset: On this dataset, except for the precision value, our GCFormer outperforms the SOTA methods in terms of the other four evaluation metrics. $F1$, IoU,

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE COMPARISON RESULTS USING DIFFERENT CD METHODS ON THE DSIFN-CD DATASET

Method	Year	Architecture	F1	IoU	OA	Pre	Rec
DTCDSCN [11]	2020	SE-ResNet	63.72	46.76	84.91	53.87	77.99
STANet[[12]	2020	ResNet18	64.59	47.66	88.49	67.71	61.68
IFNet [26]	2020	VGG16	60.10	42.96	87.83	67.86	53.94
SNUNet-CD [14]	2021	Siamese ConvNet	66.18	49.45	87.34	60.60	72.89
BIT [5]	2022	Transformer +ResNet18	69.26	52.97	89.41	68.36	70.18
ChangeFormer [18]	2022	Transformer	86.67	76.48	95.56	88.48	84.94
GCFormer	2022	Transformer+ResNet18	87.66	78.04	95.79	87.31	88.02

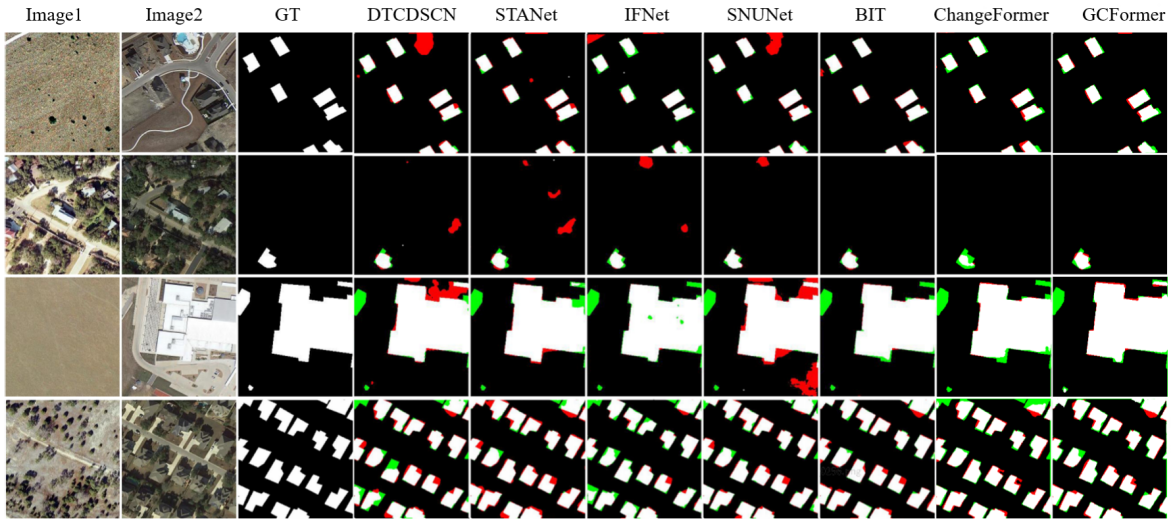


Fig. 5. Visualized results of different RSICD methods on the LEVIR-CD dataset.

recall, and OA values reach 90.83%, 83.21%, 99.08%, and 92.50%, respectively. Although the IFNet method achieves the outstanding precision value of 94.02%, its recall value is 9.57% lower than that of our GCFormer, indicating that IFNet is likely to result in missing detection. Therefore, the comprehensive $F1$ score is only 88.13%, 2.7% lower than that of our GCFormer.

2) WHU-CD Dataset: On this dataset, our GCFormer achieves the optimal results in both OA and recall values, reaching 99.29% and 90.15%, respectively. Although the H-TransCD method outperforms GCFormer slightly in $F1$ and IoU values by 0.25% and 0.42%, respectively, its recall value is 1.42% lower than that of our GCFormer. It means that H-TransCD cannot preserve the integrity of CD regions. Moreover, the IFNet method achieves a high-precision value of 96.91%, but its recall value is only 73.19%, resulting in a lower $F1$ score of 83.40%, 7.75% lower than that of our GCFormer.

3) DSIFN-CD Dataset: On this dataset, in addition to the precision value, our GCFormer achieves the optimal results in terms of other four evaluation metrics. $F1$, IoU, recall, and OA values reach 87.66%, 78.04%, 95.79%, and 88.02%, respectively. Compared with the ChangeFormer, the precision value of our GCFormer is 1.17% lower, but recall and $F1$ values are 3.08% and 0.99% higher. These results indicate that our GCFormer can achieve better overall detection performance compared with the ChangeFormer.

In order to demonstrate the detection performance of GCFormer more intuitively, the CD results are visualized in Figs. 5–7, showing some example images from the three datasets. The first and second columns of the figure display two input images from different periods, and the GT is shown in the third column. The detection results are highlighted with different colors, where white, black, red, and green represent the true positives, true negatives, false positives, and false negatives, respectively. From the first row of Fig. 5 and the first and second rows of Fig. 7, it can be seen that GCFormer performs well in the presence of "pseudo-changes" in images and can effectively reduce the occurrence of false alarms. The color difference between the before-and-after change images in the second row of Fig. 6 has limited impact on the detection performance of GCFormer.

Although the change regions in the bottom-left corner of the third row of Fig. 7 are not easily observed, and the small regions in the bottom-left corner of the first row of Fig. 6 are prone to be missed, GCFormer succeeds in detecting such subtle changes. The second row of Fig. 6 and the fourth row of Fig. 7 also verify that GCFormer can effectively reduce the phenomenon of missed detection.

The results in the fourth row of Fig. 5, the third and fourth rows of Fig. 6, and the third row of Fig. 7 demonstrate that, GCFormer can accurately detect small change regions in complex ground scenes. Overall, the visualized results clearly indicate the superiority of GCFormer in RSICD tasks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

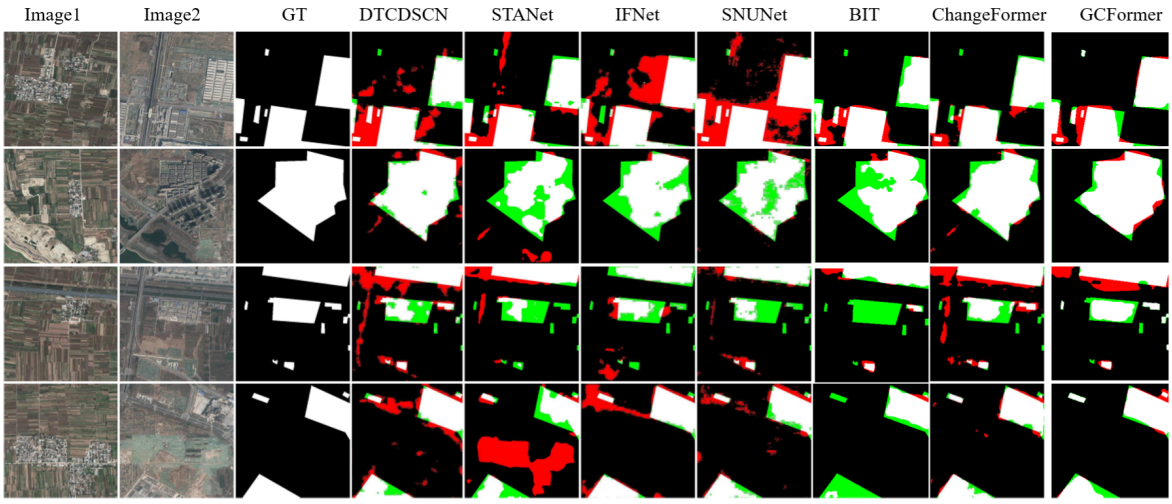


Fig. 6. Visualized results of different RSICD methods on the DSIFN-CD dataset.

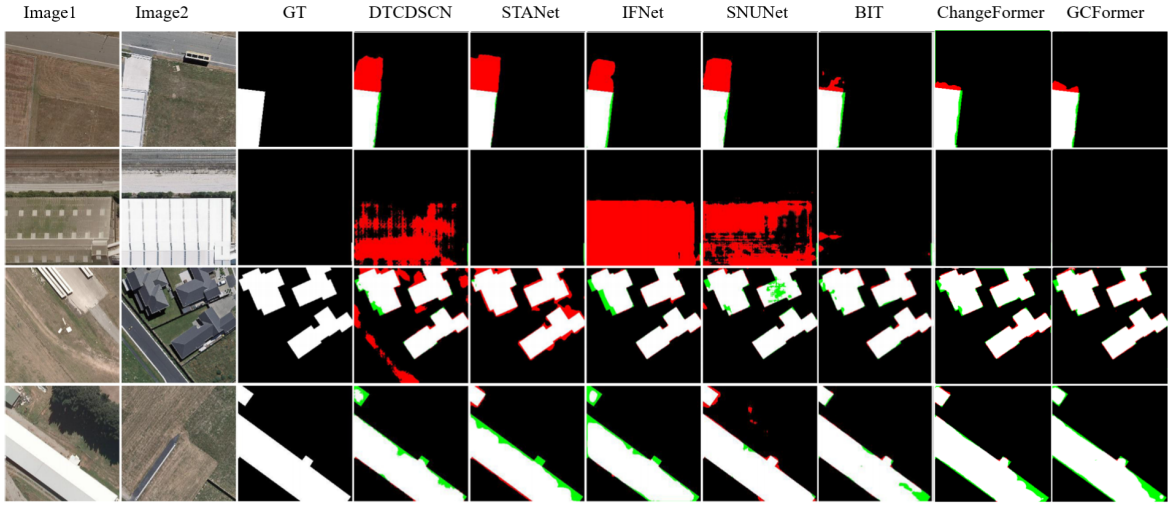


Fig. 7. Visualized results of different RSICD methods on the WHU-CD dataset.

In summary, the quantitative and qualitative comparison results comprehensively demonstrate the effectiveness of our GCFormer. Especially, for small change regions, it can significantly mitigate the occurrence of missed detection.

D. Comparison of Computational Complexity and Model Parameters

In order to compare the computational efficiency of RSICD methods, we analyzed the parameter scale (Params.) and floating-point operations per second (FLOPs) of different methods. As can be seen from Table IV and Fig. 8, compared with CNN-based CD methods, such as SNUNet and STANet, the proposed GCFormer has larger number of parameters due to its hybrid CNN + Transformer architecture. However, compared with other Transformer-based CD methods, such as ChangeFormer, H-TransCD, and IDET, our GCFormer has the lowest model parameters and computational complexity. The model parameters are only 19.27% of H-TransCD, 29.74% of IDET, and 86.35% of ChangeFormer. The computational complexity is only 11.13% of ChangeFormer, 16.72% of H-TransCD, and 18.01% of IDET. Notably, all methods in the table use the same input image size of 256 × 256.

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY AND MODEL PARAMETERS WITH OTHER METHODS

Method	Params. (M)	FLOPs (G)	F1	IoU	OA	Pre	Rec
STANet	16.93	**6.58**	87.26	77.40	98.66	83.81	**91.00**
SNUNet	12.03	27.44	88.16	78.83	98.82	89.18	87.17
BIT	**3.55**	10.60	89.31	80.68	98.92	89.24	89.37
IDET	119.12	62.76	88.90	80.20	98.10	91.30	86.60
H-TransCD	183.83	67.58	90.06	81.92	99.00	91.45	88.72
ChangeFormer	41.03	101.57	90.40	82.48	99.04	92.05	88.80
GCFormer	35.43	11.30	**90.83**	**83.21**	**99.08**	**92.50**	89.34

In general, our GCFormer can achieve a superior detection performance without sacrificing the computational efficiency, making it a more practical solution for RSICD tasks.

E. Ablation Study

To validate the effectiveness of each component in our proposed GCFormer, we conducted ablation studies on three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

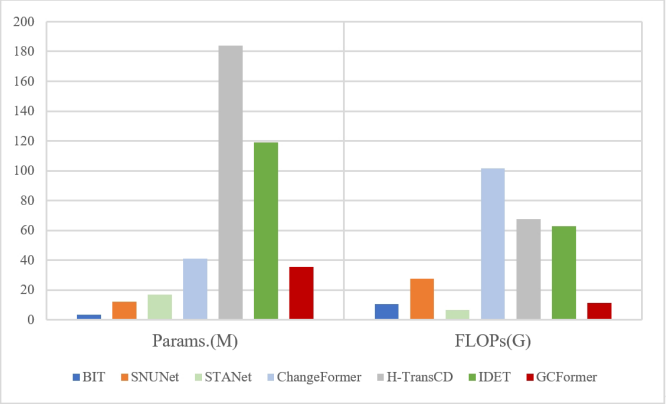


Fig. 8. Comparison of computational complexity and model parameters with other methods.

TABLE V
ABLATION EXPERIMENTAL RESULTS ON THE LEVIR-CD DATASET

Baseline	MRFCA	CRPE	F1	IoU	Pre	Rec	OA
√			89.54	81.06	90.72	88.39	98.95
√	√		90.30	82.33	91.58	89.11	99.03
√		√	90.34	82.39	**92.65**	88.16	99.05
√	√	√	**90.83**	**83.21**	89.34	**92.50**	**99.08**

TABLE VI
ABLATION EXPERIMENTAL RESULTS ON THE WHU-CD DATASET

Baseline	MRFCA	CRPE	F1	IoU	Pre	Rec	OA
√			88.64	79.60	89.24	88.04	99.10
√	√		89.44	80.91	88.73	**90.18**	99.15
√		√	89.64	81.23	89.82	89.50	99.18
√	√	√	**90.97**	**83.43**	**91.81**	90.15	**99.29**

datasets, as shown in Tables V–VII, in which the baseline represents the basic CNN + Transformer hybrid architecture that employs ResNet18 as the backbone and adopts the APE mode in Transformer.

It can be observed that, the detection performance is improved by embedding the MRFCA module into the highest layer of ResNet18 and CRPE into Transformer. Also, the MPFCA module contributes the most to the improvement of the recall value, 0.72%, 2.14%, and 3.97% over the baseline on the three datasets, respectively. This improvement is mainly attributed to the ability of MRFCA to extract multireceptive-field global context information, thus accurately locating the change regions with different scales, especially for scattered small objects.

In addition, CRPE mode incorporates more spatial position information to enhance the global context information. Therefore, it improves the detection accuracy, with the performance gains of 1.93%, 0.58%, and 1.01% on the three datasets, respectively.

Figs. 9–11 present the visualized detection results before and after incorporating MRFCA module and CRPE mode into the baseline on three different datasets, respectively. The first and second columns of the figure display two input images from different periods, and the GT is shown in the third column. The detection results are highlighted with different colors, where white, black, red, and green represent the true positives, true negatives, false positives, and false negatives, respectively.

It can be observed that, before using MRFCA, there are some missed detections and incomplete detections in small change regions, Conversely, after using MRFCA, the number of missed detections is notably reduced.

After incorporating CRPE mode, false detections have been mitigated. This is due to the network's ability to comprehensively consider the relationship between the change regions and backgrounds with the aid of more global context information, resulting in more accurate detection of the change regions, thus improving the detection accuracy of RSICD tasks.

TABLE VII
ABLATION EXPERIMENTAL RESULTS ON THE DSIFN-CD DATASET

Baseline	MRFCA	CRPE	F1	IoU	Pre	Rec	OA
√			85.31	74.38	89.12	81.81	95.21
√	√		86.54	76.28	87.44	85.78	95.47
√		√	86.62	76.40	**90.13**	83.40	95.62
√	√	√	**87.66**	**78.04**	87.31	**88.02**	**95.79**

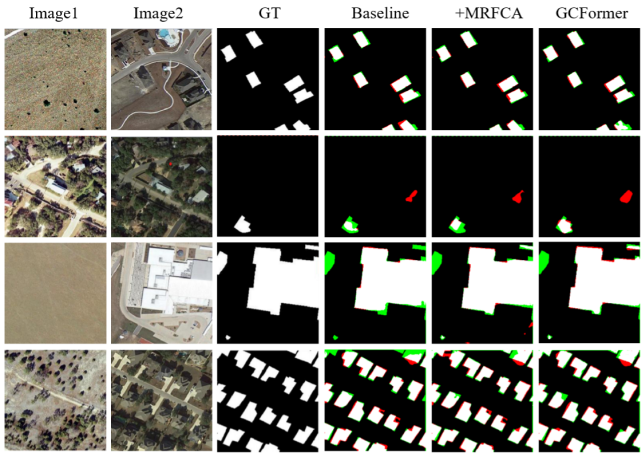


Fig. 9. Visualized results after incorporating different modules on the LEVIR-CD dataset.

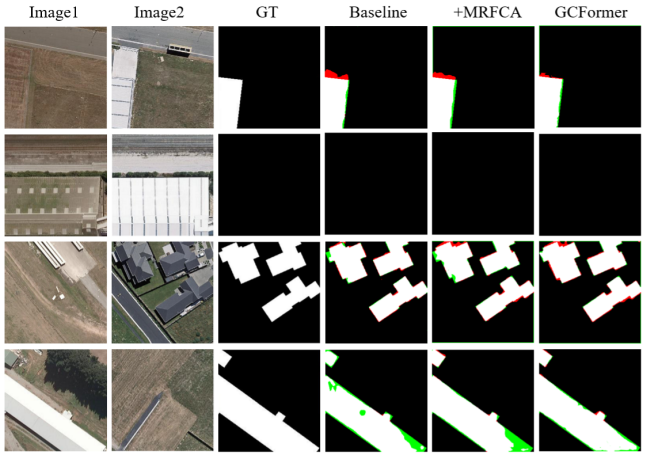


Fig. 10. Visualized results after incorporating different modules on the WHU-CD dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

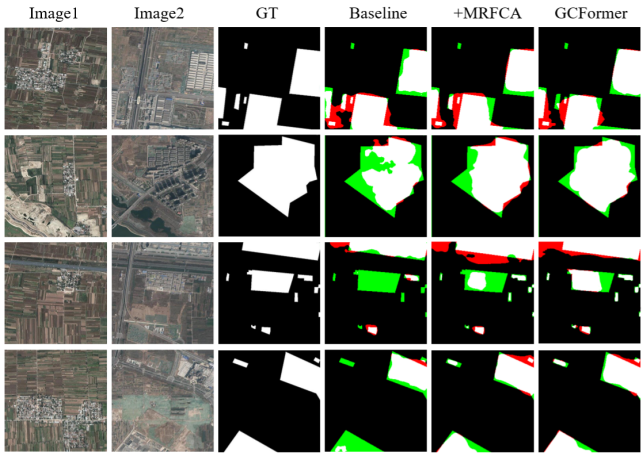


Fig. 11. Visualized results after incorporating different modules on the DSIFN-CD dataset.

TABLE VIII
EFFECT OF DIFFERENT DILATION RATES ON THE WHU-CD DATASET

Dilation Rate	F1	IoU	Pre	Rec	OA	Params. (M)	FLOPs (G)
Baseline	89.64	81.23	89.82	89.50	99.18	**34.956**	**10.594**
r=1, 2, 4	90.52	82.69	91.33	89.73	99.26		
r=1, 3, 5	90.16	82.08	91.41	88.94	99.22	35.336	11.071
r=3, 5, 7	90.09	81.97	91.20	89.01	99.22		
r=4, 8, 12, 16	89.58	81.14	91.00	88.22	99.19		
r=3, 5, 7, 9	89.99	81.80	90.20	89.77	99.21	35.429	11.298
r=2, 4, 6, 8	90.23	82.20	91.30	89.18	99.23		
r=1, 2, 4, 6	90.71	83.01	**91.98**	89.49	99.27	35.429	11.298
r=1, 3, 5, 7	**90.97**	**83.43**	91.81	**90.15**	**99.29**		

F. Effect of Different Dilation Rates on Detection Performance

To further investigate the effect of different parameter settings in MRFCA on the detection performance, we compared the performance of different dilation rates. The results on the WHU-CD dataset are shown in Table VIII. The baseline refers to using 3×3, 5×5, and 7×7 convolution operations in ResNet18 and incorporating CRPE mode to encode positional representation for Transformer. It can be seen that replacing dilated convolution operation with 3×3, 5×5, and 7×7 convolution operations leads to a performance degradation. Among the different dilation rates evaluated, the best results can be obtained with the dilation rates of 1, 3, 5, and 7, which achieves a good balance between precision metric and recall metric. Moreover, the performance indicators actually decrease, as the dilation rates increase. Based on these findings, our GCFormer sets the dilation rates to 1, 3, 5, and 7 for optimal performance.

G. Effect of the DCMR and Conv-Attention Mechanism on Detection Performance

To validate the effectiveness of DCMR and Conv-Attention mechanism in the MRFCA, we conducted the comparison experiments. The experimental results on the WHU-CD dataset are shown in Table IX. The baseline adopts the basic CNN + Transformer framework, and the proposed CRPE mode is used in Transformer.

TABLE IX
EFFECT OF DCMR AND CONV-ATTENTION ON DETECTION PERFORMANCE ON THE WHU-CD DATASET

Baseline	DCMR	Conv-Attention	Self-Attention	F1	IoU	Pre	Rec	OA
√				89.64	81.23	89.82	89.50	99.18
√	√			90.52	82.69	91.33	89.73	99.26
√		√		90.54	82.72	90.90	**90.18**	99.25
√	√	√		**90.97**	**83.43**	**91.81**	90.15	**99.29**
√	√		√	90.75	83.06	91.44	90.07	99.27

It can be observed that DCMR can improve all metrics, compared with the baseline. Specifically, $F1$, IoU, precision, recall, and OA values increase by 0.88%, 1.46%, 1.51%, 0.23%, and 0.08%, respectively. Similarly, the Conv-Attention mechanism also can significantly improve all metrics. Specifically, compared with baseline, $F1$, IoU, precision, recall, and OA values increase by 0.90%, 1.49%, 1.08%, 0.68%, and 0.07%, respectively. When the two components are both used, all metrics are further improved. Specifically, $F1$, IoU, precision, recall, and OA values reach 90.97%, 83.43%, 91.81%, 90.15%, and 99.29%, respectively.

To verify the superiority of Conv-Attention mechanism over self-attention mechanism, we also conducted the comparison experiments, as shown in Table IX. It can be seen that, compared with self-attention, Conv-Attention can obtain better detection accuracy. $F1$, IoU, precision, recall, and OA values increase by 0.43%, 1.23%, 0.91%, 0.08%, and 0.02%, respectively.

In summary, both of DCMR and Conv-Attention mechanism can effectively enhance the global context information. Their combination can further strengthen the ability of the network, thus improving the detection performance.

H. Effect of Different Position Encoding Modes on CD Performance

Considering the impact of the query vector on the final detection performance of RSICD tasks, different position encoding modes are compared. The comparison results are shown in Table X. It can be seen that due to the reason that the context mode changes the encoding manner, the proposed CRPE can achieve a higher detection performance than APE.

To further verify the effect of different similarity measurement methods on the detection performance, both dot product and cosine function are used to measure the similarity between tokens in the self-attention mechanism. The comparison results are shown in Table XI. It can be observed that, when CRPE mode used, the cosine measurement method performs better than the dot product operation.

I. Effect of Transformer Parameters Configuration on CD Performance

1) Token Length: When performing encoding and decoding operations, the length of tokens is an important parameter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

TABLE X
EFFECT OF DIFFERENT POSITION ENCODING MODES ON THE WHU-CD DATASET

Position Encoding Mode	F1	IoU	Pre	Rec	OA	Params. (M)	FLOPs (G)
APE	89.44	80.91	88.73	**90.18**	99.15	**35.395**	**11.191**
CRPE	**90.97**	**83.43**	**91.81**	90.15	**99.29**	35.429	11.298

TABLE XI
EFFECT OF DIFFERENT SIMILARITY MEASUREMENT METHODS ON THE WHU-CD DATASET

Position Encoding Mode	Measurement Method	F1	IoU	Pre	Rec	OA
CRPE	**Cosine**	**90.97**	**83.43**	**91.81**	90.15	**99.29**
	Dot product	90.54	82.72	90.90	**90.18**	99.25

TABLE XII
EFFECT OF DIFFERENT TOKEN LENGTHS ON THE WHU-CD DATASET

Token Length	F1	IoU	Pre	Rec	OA	Params. (M)	FLOPs (G)
32	88.64	79.60	89.24	88.04	99.10	35.438	11.324
16	90.69	82.97	**92.16**	89.27	99.27	35.433	11.309
8	90.91	83.34	91.91	89.94	99.29	35.430	11.301
4	**90.97**	**83.43**	91.81	**90.15**	**99.29**	35.429	11.298
2	90.35	82.40	91.17	89.54	99.24	**35.428**	**11.296**

The impact of different token lengths on detection accuracy is tested in this study, as shown in Table XII. It can be seen that, when the token length is reduced from 32 to 4, there is a significant improvement in all evaluation metrics. This indicates that compact tokens are more capable of representing change semantic information, whereas longer tokens make the dependency between each element more complicated, thereby affecting the detection performance. However, when the token length is further reduced from 4 to 2, the detection performance declines. This is because when the token length is too short, the model may lose some essential information related to change regions, thereby affecting the detection accuracy. Therefore, in this article, token length is set to 4, so as to achieve a balance between the detection accuracy and the computational efficiency.

2) Encoder and Decoder Depth: The depths of the encoder and decoder are also important parameters of Transformer. In this article, experiments are conducted with different depths of encoder and decoder, as shown in Table XIII. It can be seen that when the depth of the encoder is increased, there is no significant improvement in $F1$, IoU, Pre, Rec, and OA metrics. This means that a single-layer encoder is sufficient for learning the relationship between tokens. However, it can also be seen that the detection performance is roughly positively correlated with the depth of the decoder. This is because more context-rich tokens are obtained after each layer of the decoder, which is conducive to more efficiently modeling long-range dependency.

TABLE XIII
EFFECT OF THE DEPTHS OF THE ENCODER AND DECODER ON THE WHU-CD DATASET

Encoder Depth	Decoder Depth	F1	IoU	Pre	Rec	OA	Params (M)	FLOPs (G)
1	1	89.91	81.66	90.28	89.53	99.20	**30.242**	**9.182**
	4	90.51	82.72	90.90	90.18	99.25	32.638	10.089
	8	**90.97**	**83.43**	91.81	90.15	**99.29**	35.429	11.298
	16	90.68	82.95	91.45	89.92	99.27	41.009	13.716
2	8	90.80	83.16	**92.79**	88.90	99.29	36.126	11.298
	16	90.49	82.63	91.45	89.54	99.25	41.707	13.716
4	8	90.09	81.97	91.77	88.47	99.23	37.521	11.299
	16	89.96	81.75	90.66	81.75	99.21	43.102	13.717
8	8	90.65	82.91	90.11	**91.20**	99.25	40.312	11.302
	16	89.74	81.39	90.38	89.11	99.19	45.890	13.720

Increasing the depth of the decoder from 8 to 16 leads to the decrease in detection accuracy, while significantly increasing computational complexity and model parameters. To achieve a good balance between the detection efficiency and the accuracy, our GCFormer sets the depths of the encoder and decoder to 1 and 8, respectively.

V. CONCLUSION

The proposed GCFormer is designed to address the limitations of currently existing RSICD methods, particularly on detecting small scattered objects that often suffer from missed detections. The ability of capturing global context information of network is enhanced through designed MRFCA, which can enhance multireceptive-field global context information from the highest layer of CNN, and CRPE mode, which enhances the extraction and representation ability of global semantic features. This article presents a thorough analysis of the impact of different parameters on the detection performance. The experimental results demonstrate that the GCFormer can efficiently detect the small change regions, thus improving the detection accuracy. Overall, GCFormer offers a promising solution for RSICD tasks.

It is worth noting that the proposed method achieves superior detection performance on three public datasets of LEVID-CD, WHU-CD, and DSIFN-CD. However, it does have higher computational complexity compared with CNN-based RSICD methods. In future work, it would be valuable to reduce the model complexity while maintaining high detection accuracy. This requires research into specific techniques, such as model compression or designing efficient network architecture.

REFERENCES

[1] R. J. Radke, S. Andra, O. Al-Kofahi, and B. Roysam, "Image change detection algorithms: A systematic survey," *IEEE Trans. Image Process.*, vol. 14, no. 3, pp. 294–307, Mar. 2005, doi: 10.1109/TIP.2004.838698.

[2] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, Long Beach, CA, USA, 2017, pp. 5998–6008.

[3] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K., 2020, pp. 213–229.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.

[4] Q. Guo, J. Zhang, S. Zhu, C. Zhong, and Y. Zhang, "Deep multiscale Siamese network with parallel convolutional structure and self-attention for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406512, doi: 10.1109/TGRS.2021.3131993.

[5] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514, doi: 10.1109/TGRS.2021.3095166.

[6] X. Hu, P. Zhang, and Q. Zhang, "A novel framework of CNN integrated with AdaBoost for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Waikoloa, HI, USA, Sep. 2020, pp. 2643–2646, doi: 10.1109/IGARSS39084.2020.9324261.

[7] A. Ma et al., "Fast sequential feature extraction for recurrent neural network-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5920–5937, Oct. 2021, doi: 10.1109/TGRS.2020.3018449.

[8] S. Sun, L. Mu, L. Wang, and P. Liu, "L-UNet: An LSTM network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3041530.

[9] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 4063–4067, doi: 10.1109/ICIP.2018.8451652.

[10] X. Qu, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Change detection in synthetic aperture radar images using a dual-domain network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4013405, doi: 10.1109/LGRS.2021.3073900.

[11] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[12] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[13] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021, doi: 10.1109/JSTARS.2020.3037893.

[14] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8007805, doi: 10.1109/LGRS.2021.3056416.

[15] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 548–558, doi: 10.1109/ICCV48922.2021.00061.

[16] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 538–547, doi: 10.1109/ICCV48922.2021.00060.

[17] X. Pan et al., "On the integration of self-attention and convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 805–815, doi: 10.1109/CVPR52688.2022.00089.

[18] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Kuala Lumpur, Malaysia, 2022, pp. 207–210, doi: 10.1109/IGARSS46834.2022.9883686.

[19] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[20] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002, doi: 10.1109/ICCV48922.2021.00986.

[21] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713, doi: 10.1109/TGRS.2022.3160007.

[22] Q. Ke and P. Zhang, "Hybrid-TransCD: A hybrid transformer remote sensing image change detection network via token aggregation," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 4, p. 263, Apr. 2022.

[23] Q. Guo et al., "IDET: Iterative difference-enhanced transformers for high-quality change detection," 2022, *arXiv:2207.0924*.

[24] F. Song, S. Zhang, T. Lei, Y. Song, and Z. Peng, "MSTDSNet-CD: Multiscale Swin transformer and deeply supervised network for change detection of the fast-growing urban regions," *IEEE Trans. Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art no. 6508505, doi: 10.1109/LGRS.2022.3165885.

[25] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019, doi: 10.1109/TGRS.2018.2858817.

[26] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

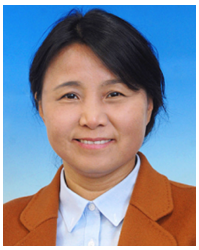
[27] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2009, doi: 10.1109/TNN.2008.2005605.

[28] S. Saha, L. Mou, X. X. Zhu, F. Bovolo, and L. Bruzzone, "Semisupervised change detection using graph convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 607–611, Apr. 2021, doi: 10.1109/LGRS.2020.2985340.

[29] J. Wu, B. Li, Y. Qin, W. Ni, H. Zhang, and Y. Sun, "A multiscale graph convolutional network for change detection in homogeneous and heterogeneous remote sensing images," 2021, *arXiv:2102.08041*.

Wanting Yu currently pursuing the M.S. degree with the Faculty of Information Technology, Beijing University of Technology, Beijing, China.

She is with Beijing Key Laboratory of Computational Intelligence and Intelligent System, Beijing University of Technology. Her research interests include remote image processing.



Li Zhuo (Member, IEEE) received the Ph.D. degree in pattern recognition and intelligent system from Beijing University of Technology, Beijing, China, in 2004.

She is currently a Professor with the Faculty of Information Technology, Beijing University of Technology. She is also with the Beijing Key Laboratory of Computational Intelligence and Intelligent System, Beijing University of Technology. She has authored or coauthored more than 180 refereed journals and conference papers and has written six book chapters. Her research interests include image/video processing, image recognition, and medical image processing.

Jiafeng Li (Member, IEEE) received the Ph.D. degree in pattern recognition and intelligence system from Beihang University, Beijing, China, in 2016.

He was with the Department of Neurosurgery, University of Pittsburgh, Pittsburgh, PA, USA, as a Visiting Scholar, from 2014 to 2015. He is currently an Assistant Professor with the Faculty of Information Technology, Beijing University of Technology, Beijing. He is also with Beijing Key Laboratory of Computational Intelligence and Intelligent System, Beijing University of Technology. His research interests include image restoration/enhancement, object detection, and object tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:16:36 UTC from IEEE Xplore. Restrictions apply.